An official website of the United States government Here's how you know
National Institutes of Health / National Library of Medicine

URL of this page: https://medlineplus.gov/cpr.html

CPR

Also called: Cardiopulmonary resuscitation

When someone's blood flow or breathing stops, seconds count. Permanent brain damage or death can happen quickly. If you know how to perform cardiopulmonary resuscitation (CPR), you could save a life. CPR is an emergency procedure for a person whose heart has stopped (called sudden cardiac arrest [https://medlineplus.gov/suddencardiacarrest.html]) or who is no longer breathing. CPR can maintain the blood flow breathing until emergency medical help [https://medlineplus.gov/emergencymedicalservices.html] arrives.

Take these steps if someone is in sudden cardiac arrest:

- Call 911.
- If someone else is around, have them look for an automated external defibrillator (AED). An AED is a device that sends an electric shock to the heart to try to restore its normal rhythm. AEDs are available in many public places such as schools, businesses, and airports .
- Make sure that the person with sudden cardiac arrest is lying on their back on a firm surface.
- If you haven't had any CPR training, you can do "hands-only" CPR for a teen or adult whose heart has stopped beating ("hands-only" CPR isn't recommended for children). "Hands-only" CPR uses chest compressions to keep blood circulating. Chest compressions involve pushing hard and fast in the center of the chest.
- If you've had training, you can use chest compressions, clear the airway, and do rescue breathing. Rescue breathing helps get oxygen to the lungs for a person who has stopped breathing. To keep your skills up, repeat the CPR training every two years.
- Use the AED as soon as possible, if there is one available.

Start Here

- Cardiopulmonary Resuscitation (CPR): First Aid [https://www.mayoclinic.org/first-aid/first-aid-cpr/basics/art-20056600?p=1] (Mayo Foundation for Medical Education and Research)
  Also in Spanish [https://www.mayoclinic.org/es/first-aid/first-aid-cpr/basics/art-20056600?p=1]
- What Is CPR? [https://cpr.heart.org/en/resources/what-is-cpr] (American Heart Association)

Specifics

- Empowered to Keep the Beat: Hands-Only CPR [https://www.empoweredtoserve.org/en/community-resources/health-lessons/hands-only-cpr] (American Heart Association)
- Hands-Only CPR [https://cpr.heart.org/en/cpr-courses-and-kits/hands-only-cpr] (American Heart Association)

Images

- CPR - adult - series [https://medlineplus.gov/ency/presentations/100219_1.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/esp_presentations/100219_1.htm]
- CPR - child 1 to 8 years old - series -- Check for responsiveness [https://medlineplus.gov/ency/presentations/100215_1.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/esp_presentations/100215_1.htm]
- CPR - infant - series [https://medlineplus.gov/ency/presentations/100216_1.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/esp_presentations/100216_1.htm]

Videos and Tutorials

- How to Do Compression-Only CPR [https://www.youtube.com/watch?v=VZqG-tcZvfE] (American Red Cross)
- How to Give CPR to a Child [https://www.youtube.com/watch?v=lXK7IjyhbL0] (American Heart Association)
- How to Perform Hands-Only CPR [https://www.youtube.com/watch?v=6eRwgM2Pa4o] (American Red Cross)
- How to Use an AED on an Adult [https://www.youtube.com/watch?v=in8j2Q2z3HE] (American Red Cross)
- How to Use an AED on Children and Infants [https://www.youtube.com/watch?v=yp5gjJa2FxA] (American Red Cross)
- Using an AED on a Pregnant Person [https://www.youtube.com/watch?v=YCEywOo6Wbg] (American Red Cross)
- What You Should Do Before Giving CPR [https://www.youtube.com/watch?v=ytMPHjwfcrI] (American Red Cross)

Statistics and Research

- CPR Facts and Stats [https://cpr.heart.org/en/resources/cpr-facts-and-stats] (American Heart Association)

Clinical Trials

- ClinicalTrials.gov: Cardiopulmonary Resuscitation [https://clinicaltrials.gov/search?intr=cardiopulmonary+resuscitation&aggFilters=status:not%20rec] (National Institutes of Health)

Journal Articles

References and abstracts from MEDLINE/PubMed (National Library of Medicine)

- Article: Effect of low-dose, high-frequency advanced life support training versus annual full-day... [https://www.ncbi.nlm.nih.gov/pubmed/42323602]
- Article: Effect of a Mobile App on Medication Errors During Simulated Pediatric... [https://www.ncbi.nlm.nih.gov/pubmed/42319567]
- Article: Leveraging immersive technology with virtual emulation for training of nonmedico combatants:... [https://www.ncbi.nlm.nih.gov/pubmed/42301684]
- CPR -- see more articles [https://pubmed.ncbi.nlm.nih.gov/?term=%22Cardiopulmonary+Resuscitation%22%5Bmajr%3Anoexp%5D+AND+humans%5Bmh%5D+AND+english%5Bla%5D+AND+%22last+1+Year%22+%5Bedat%5D+AND+%28patient+education+handout%5Bpt%5D+OR+guideline%5Bpt%5D+OR+clinical+trial%5Bpt%5D%29++NOT+%28letter%5Bpt%5D+OR+case+reports%5Bpt%5D+OR+editorial%5Bpt%5D+OR+comment%5Bpt%5D%29+AND+free+full+text%5Bsb%5D+]

Find an Expert

- American Heart Association [https://www.heart.org/en/]
- American Red Cross [https://www.redcross.org/]
  Also in Spanish [https://www.redcross.org/cruz-roja.html]
- National Heart, Lung, and Blood Institute [https://www.nhlbi.nih.gov/] NIH

Children

- CPR (For Parents) [https://kidshealth.org/en/parents/cpr.html] (Nemours Foundation)
  Also in Spanish [https://kidshealth.org/es/parents/cpr.html]
- CPR: A Real Lifesaver (For Kids) [https://kidshealth.org/en/kids/cpr.html] (Nemours Foundation)
  Also in Spanish [https://kidshealth.org/es/kids/cpr.html]



MEDICAL ENCYCLOPEDIA

Breathing difficulties - first aid [https://medlineplus.gov/ency/article/000007.htm]

CPR [https://medlineplus.gov/ency/article/000010.htm]

CPR - adult - series [https://medlineplus.gov/ency/presentations/100219_1.htm]

CPR - adult and child after onset of puberty [https://medlineplus.gov/ency/article/000013.htm]

CPR - child 1 to 8 years old - series -- Check for responsiveness [https://medlineplus.gov/ency/presentations/100215_1.htm]

CPR - infant [https://medlineplus.gov/ency/article/000011.htm]

CPR - infant - series [https://medlineplus.gov/ency/presentations/100216_1.htm]

CPR - young child (age 1 year to onset of puberty) [https://medlineplus.gov/ency/article/000012.htm]

Myocardial contusion [https://medlineplus.gov/ency/article/000202.htm]

Related Health Topics

First Aid [https://medlineplus.gov/firstaid.html]

Sudden Cardiac Arrest [https://medlineplus.gov/suddencardiacarrest.html]

National Institutes of Health

The primary NIH organization for research on *CPR* is the National Heart, Lung, and Blood Institute [http://www.nhlbi.nih.gov/]

MedlinePlus links to health information from the National Institutes of Health and other federal government agencies. MedlinePlus also links to health information from non-government Web sites. See our disclaimer [https://medlineplus.gov/disclaimers.html] about external links and our quality guidelines [https://medlineplus.gov/criteria.html] .

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Learn how to cite this page